

GeoChem Explorer User Guide

Mineralabs

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GeoChem Explorer User Guide

GeoChem Explorer is a desktop app for exploring drillhole geochemistry, geology, structure, collar locations, core photos, and derived analysis products from local project folders. It is built around a linked workspace: filters and selections are shared across panels, so a subset selected in one view can be inspected immediately in the others.

This guide describes the app from a user point of view: what each panel does, what the sidebar controls do, and how projects are created and maintained.

1. Main Workspace

The app opens into a four-panel workspace after a project is loaded. The default layout is:

- Scatter Plot
- Histogram
- Plan View
- Downhole

Use the top-bar Workspace menu to choose which panel appears in each of the four slots. Available panel types are:

- Scatter
- Histogram
- Plan Map
- Downhole

- Report Generator

Each panel has a maximize button in its header. Maximizing is useful for detailed work such as interpreting a stereonet, viewing a downhole heatmap, or reviewing a report packet.

2. Top Bar

The top bar shows the loaded project name, filtered row counts, selected interval counts, and the active filter engine.

Status Metrics

- Rows shows the filtered row count.
- Preview appears when the app is showing a sampled preview for a large raw-mode filter.
- Selected shows the number of selected intervals.
- Engine shows whether filtering is ready, applying, running through DuckDB, or using in-memory filtering.

Top-Bar Actions

- Clear Selection: clears the linked interval selection.
- Project Settings: opens per-project settings, including core photo library roots and project input reload access.
- App Settings: opens global computer-level settings such as update delivery, default project folders, default input source folders, and default core photo roots.
- Project Inputs: returns to the project loader so source files can be previewed, synced, and reloaded.
- Check Updates: checks the public GeoChem Explorer GitHub releases, and any configured internal release folders, for app updates.
- Workspace: changes which panels appear in the four workspace slots.
- Export: exports selected intervals, filtered intervals, or a filtered hole list to CSV.
- Save Session / Load Session: saves or restores project path, active filters, selected rows, analytes, panel layout, PCA/clustering state, and other workspace UI state.

3. Sidebar Overview

The sidebar controls the shared project state. Most views respond immediately to sidebar filters and analyte choices.

Sidebar sections:

- Filters
- Analytes
- Tools
- QA Summary

The sidebar can also open modal windows for saved subsets, date phases, lithology colours, analyte colouring, PCA loadings, cluster profiles, and QA issue tables.

4. Filters Sidebar

Interval Mode

Use the Intervals selector to choose the working interval table:

- Raw assays

- 2m composite
- 5m composite

Composites are useful when you want equal-length samples for PCA, clustering, or cleaner spatial comparisons.

Top-Level Filters

The top filter group filters the whole app by:

- Prospect
- Dataset
- Major Split
- Hole Type
- Weathering

These controls are disabled or hidden when the loaded project does not contain the relevant field.

Saved Filters

Subset Data stores and reapplies reusable named hole subsets. A subset stores:

- Hole IDs
- Prospect context
- Interval mode
- Top-level filters such as prospect, dataset, hole type, weathering, lith codes, and depth range

Use `New Subset...` to save a current hole set. The app saves holes from:

1. Explicit Hole ID filter, if present.
2. Currently selected intervals, if present.
3. Current filtered hole set, if no explicit hole IDs or selected intervals exist.

This order matters. If you rectangle-select holes on the plan map, click `Use selected or visible holes`, then save a subset, the subset should represent the selected holes.

Date Phase stores reusable date-based hole filters from collar date fields. A date phase can be:

- Between
- On
- Before
- After

Saved subsets and date phases persist outside sessions in the project cache.

Advanced Hole Filters

The Hole ID section contains:

- Find holes by Starts With, Contains, Ends With, or Exact Match.
- Select Matches to replace the current hole filter.
- Add Matches to add matches to the current hole filter.
- Direct hole ID entry with Add or Replace.
- Manual multi-select hole list.
- Depth Min and Depth Max.

The Lith Code section contains:

- Lithology presets: Custom, ML - Most Used, Intermed / Mafic, Felsic, and Common Metaseds.
- Manual lith code multi-select.
- Assign Colours... to set project-default lith colours.

Reset Filters

Reset Filters clears filters back to the project-wide dataset. It also clears active saved subset and date phase state.

5. Analytes Sidebar

The Analytes section controls the analytes used by several panels.

Main Analyte Selectors

- Scatter X Analyte
- Scatter Y Analyte
- Colour Analyte
- Histogram Analyte

Ternary mode uses Scatter X, Scatter Y, and Colour as the A/B/C ternary axes.

If PCA has been run, PCA score fields are also available in the analyte selectors.

Analyte Colouring

Analyte Colouring... opens project-level display scale tools. These allow custom numeric colour presets for analytes, including discrete binning and continuous colour scales. Colour presets can be assigned back to analytes and are reused by charts that colour by that analyte.

The same presets are used for drillhole assays and surface samples. For example, you can create an Au soils preset, then choose it in Plan View with Surface Scale > Saved Preset when the surface analyte is Au. Plan View also has a Drill Scale selector, so drillhole assay bars and surface samples can use different saved presets even when they use the same analyte.

6. Tools Sidebar

Scatter / Histogram Tools

- Clip Lower % and Clip Upper %: trim extreme values for scatter and histogram display.
- Reset Clip: restores full percentile range.
- Log X Axis and Log Y Axis: log-scale scatter axes and relevant downhole views.

Scattergram Pairs

The app can suggest useful scattergram pairs.

Modes:

- Trend: favours high-correlation pairs.
- Contrast: favours pairs that may separate domains or show differences.

Scopes:

- Filtered
- All Data

Clicking a suggestion sets the scatter X/Y analytes and opens the scatter view.

Ternary Triplets

The app can suggest ternary combinations from presets and data coverage. Suggestions show overlap counts and edge ratios. Clicking a suggestion sets the three ternary axes and opens ternary mode.

Downhole Intercepts

Configure intercept reporting before using Generate intercepts/stats in the Downhole panel.

Settings:

- Primary Analyte
- Secondary Analyte
- Cut-off (ppm)
- Max Internal Dilution (m)
- Exclude < m
- Report Primary In percent or ppm
- Hatch Colour

PCA

PCA runs on the currently filtered interval set.

Controls:

- Components: number of principal components, 2 to 8.
- PCA Analytes...: choose the analytes used in PCA.
- Row Cap: optional row limit for large datasets.
- Run PCA
- Clear
- View Loadings: opens a loadings table with copyable TSV.

PCA is disabled when raw filtering is in preview mode, because PCA needs an exact working set.

After PCA runs:

- The Scatter panel can switch to PCA mode.
- PCA scores become available in analyte selectors.
- PCA loadings and explained variance are shown in the sidebar.

Clustering

Clustering groups intervals using comparable assay suites.

Controls:

- Auto K or Manual K
- Cluster Assay Suite
- Cluster Count (K) or Auto K Max
- Run Clusters
- Clear
- View Profile

- Assign Lith...

Cluster outputs can be used in:

- Scatter colouring
- Plan map colouring
- Downhole cluster/lith interpretation tracks

The cluster profile can be viewed as z-scores or raw means and copied as TSV.

QA Summary

The QA section reports:

- Missing analyte rows
- Negative analyte rows
- Duplicate intervals
- Overlapping intervals

Duplicate and overlap counts are clickable when issues exist. The modal tables show the affected holes, intervals, and IDs so the source files can be corrected.

7. Scatter / Ternary / PCA Panel

The Scatter panel has three modes:

- Scatter Plot
- Ternary Plot
- PCA Plot

Scatter Plot Mode

Scatter mode plots Scatter X Analyte against Scatter Y Analyte.

Render modes:

- Points
- Density
- Hexbin
- Contour
- Trend
- QQ plot

Optional colouring:

- By selected Colour Analyte
- By lithology
- By cluster

Selections made in the scatter plot are linked to the other panels. Clicking a point selects a row. Box/lasso selection selects multiple intervals.

Ternary Plot Mode

Ternary mode uses:

- X analyte as axis A
- Y analyte as axis B

- Colour analyte as axis C

Render modes:

- Points
- Density

Tools include:

- Concave hull display
- Axis rotation
- Colour by lith or cluster when available

PCA Plot Mode

PCA mode is available after running PCA in the sidebar.

Capabilities:

- 2D PCA plot.
- 3D PCA plot when at least 3 components exist.
- Select X/Y/Z principal components.
- Colour by prospect, lith, cluster, or continuous analyte/PCA score.

8. Histogram Panel

The Histogram panel has four modes:

- Histogram
- Correlation
- Groups
- Domain

Histogram Mode

Shows the selected Histogram Analyte distribution for the current filtered dataset. Selected intervals are overlaid where applicable. If the analyte includes non-positive values, the panel can toggle between hiding non-positive values and showing all values.

Correlation Mode

Displays a log10 positive-value correlation heatmap for analytes. Click a heatmap cell to set the Scatter panel X/Y analytes to that pair.

Groups Mode

Compares the histogram analyte by a categorical field. Grouping options depend on the project and filtered data, such as lithology, prospect, dataset, weathering, or other available group fields. A side legend shows group counts.

Domain Mode

Domain mode shows geochemical domain template plots where the required analytes are available. It supports both XY templates and ternary templates.

Tools can include:

- Domain plot preset selector.
- Colour by lith.

- Cu $\geq$ 0.5% density overlay where available.
- Domain or PCD template overlay.
- Dot opacity.

Selections in Domain mode are linked to the global selection.

9. Plan View / Section / Stereonet Panel

This panel is titled Plan View and contains three sub-modes:

- Plan
- Section
- Stereonet

Plan Mode

Plan mode maps holes and intervals in X/Y space.

Layer modes:

- Both: show drillholes and surface samples together.
- Surface: show only surface samples.
- Drill Holes: show only drill collars, traces, and assay bars.

Plan styles:

- Points
- Trace
- Trace + Assay Bar

Colour modes:

- Analyte
- Cluster, when cluster assignments are available

Other plan tools:

- Drill Scale: choose the assigned/default drill assay colour scale or a saved preset for the active drill analyte.
- Follow Drill Analyte: when enabled, surface samples use the same analyte as the drillhole colour analyte.
- Surface Analyte: choose an independent surface-sample analyte when Follow Drill Analyte is off.
- Surface Scale: choose Auto Jenks 6, Shared with Drill, or Saved Preset for the surface sample layer.
- Show derived strike ticks.
- Show all interval points when the filtered dataset is small enough.
- Hover mode: Auto, On, Off.
- Use selected or visible holes: if intervals are selected, converts selected intervals to a hole filter; otherwise uses holes visible in the current viewport.
- Clear selection.
- Clear hole filter.

Surface samples are shown as their own marker layer above drill traces. Their hover text starts with `Surface Sample` and reports the selected surface analyte value, dataset, prospect, and from/to interval if available.

By default, surface samples use `Auto Jenks 6`. This calculates Jenks natural breaks from the currently filtered surface-sample values for the selected surface analyte, using up to six bins. If there are fewer useful distinct values, the app reduces the bin count automatically. Use `Saved Preset` when you want a project-defined scale from `Analyte Colouring`, such as a company-standard soil Au or Cu scale. Use `Shared with Drill` when drillholes are visible, the surface analyte matches the drill analyte, and you want both layers to use the same scale.

In compact four-panel view, `Plan View` shows one colour scale when both drill assays and surface samples are visible. The drill assay scale is preferred. In maximized `Plan View`, drill and surface scales can be shown side by side when they are different.

Plan selections are linked to the rest of the app. Use `Plotly box/lasso` selection to select intervals, then use `Use selected` or `visible holes` to convert that selection into a hole filter that can be saved as a subset.

Section Mode

The `Section` tool lets you draw a section line on the plan. The section modal then projects nearby holes onto the line.

Section controls include:

- Front and back distance from the section line.
- Section style: points, traces, or trace + assay bar.
- Colour by analyte or cluster.
- Optional cluster bars beside assay bars.
- Show labels.
- Show strike ticks.
- Depth marker spacing.
- Step section line forward/backward.
- Redraw line.
- Flip section view.

Stereonet Mode

Stereonet mode visualizes lower-hemisphere structural or derived orientation data.

Sources:

- Structural
- Grade Continuity
- Grade Anisotropy
- Grade Strike

Structural filters:

- Structure type
- Orientation quality

Derived-grade settings:

- Analyte
- Cutoff
- Neighbour count

Display controls:

- Hover orientation as dip/dip-direction or strike/dip.
- Render as dots, contour, or both.
- Great circles on/off.

Selections in stereonet mode can update the hole filter. Shift-selection can add/toggle holes depending on interaction.

10. Downhole Panel

The Downhole panel inspects one hole or a multi-track set of assays down depth.

Core Controls

Header actions include:

- Follow/pin current selection.
- Clear selected intervals.
- Generate intercepts/stats.
- Compare holes.
- Fullscreen core image navigator.
- Element correlation matrix.
- Heatmapper.
- Toggle panel tools.

Panel tools include:

- Active hole selector.
- Single-hole analyte selector.
- Assay source status, such as lab, pXRF, mixed, or no data.
- Depth scale: auto fit, tray photo scale, or fixed pixel-per-50 m options.
- Draw mode: dot, line, line + dot, bar, lollipop, or mountain.
- Shading: none, lith, weathering, or oxidation.
- Log X toggle.
- Core photos toggle.
- Photo type: dry, wet, or row.

Single and Multi-Track Views

Single view plots one analyte down the active hole.

Multi view can show multiple tracks. The track selector module supports up to the configured track limit and can be populated from built-in or user-saved track presets.

Built-in track presets include mineral proxy, interpretation, targeting, mafic indicator, base metal, alteration driver, sericite/chlorite, pathfinder, and gold context style groupings when matching analytes exist.

Intercepts and Stats

Use sidebar Downhole Intercepts settings, then click Generate intercepts/stats in the Downhole panel. The panel reports:

- Intercept lines based on cutoff, dilution, and minimum length.
- Summary statistics for analytes in the active hole.

Hole Comparison

The comparison modal shows multiple holes side by side with a shared depth scale.

Controls include:

- Single or multi-track view.
- Hole selectors for each comparison slot.
- Shared scale, draw mode, shade mode, and log X.
- Paging through groups of four holes.

Heatmapper

The downhole heatmap modal summarizes many analytes down a hole.

Controls include:

- Suite order or clustered order.
- Fit hole or full depth.
- Export HTML.

Clustered order includes a dendrogram where enough data exists.

Element Correlation Matrix

The correlation matrix modal computes element correlations for the active hole.

Controls include:

- Suite order.
- Clustered order.

The matrix reports analytes dropped because of insufficient or flat values.

Core Image Navigator

The fullscreen core navigator can combine:

- Dry tray photos.
- Wet tray photos.
- Downhole image lanes where available.
- Computed core colour lanes.
- Lithology lanes.
- Chemistry lanes.
- Point-data lanes.
- Intercept lanes.

It supports panning, zooming, lane layout presets, custom lane widths, lane ordering, and saved layout presets. Project-level core photo roots are configured in Project Settings or during project setup.

11. Campaign Report Builder Panel

The Report Generator workspace panel builds Reaper Executive EM report material from the loaded project and configured reporting sources.

Report modes:

- Daily
- Weekly
- Campaign-to-date

Actions:

- Generate HTML Preview
- Save HTML + Packet
- Copy HTML
- Open HTML
- Open Output Folder

The panel shows:

- Scope.
- Drilled holes.
- Drilled metres.
- Planned remaining metres.
- Intercept count.
- Warning count.
- Source status table.
- Validation summary.
- Facts packet JSON preview.
- HTML email preview.

Report behaviour is configured in the project reporting section.

12. Linked Selection Behaviour

Most plots share the same selected interval IDs.

Common behaviours:

- Click an interval/point to select it.
- Box or lasso select to select many intervals.
- Double-click in some plots clears selection or clears a local hole filter.
- Selected intervals are highlighted across linked panels.
- Top bar Clear Selection clears the shared selection.

Plan and stereonet views can also turn selections into hole filters, which is useful for creating saved subsets.

13. Project Creation and Loading

Projects are ordinary folders on disk. A project usually has:

```
project-folder/  
  project.json  
  input/
```

collar.csv
survey.csv
assays.csv
surface_samples.csv
lithology.csv
geology.csv
structural.csv
magnetic_susceptibility.csv

Required datasets for drillhole projects:

- Collar
- Survey
- Assays

Surface-only projects can be loaded from a surface sample file with Sample ID, X, Y, and analyte columns.

Optional datasets:

- Surface samples such as soils, rock chips, grab samples, or other point geochemistry
- Lithology
- Geology
- Structural
- Magnetic susceptibility
- Secondary assay sources such as pXRF
- Custom interval or point data
- Core photo manifests
- Reporting source workbooks

Loading an Existing Project

Use **Select Project Folder** in the Project Loader. The app reads:

- project.json
- Input files from the input folder
- Cached project data from .geochem-cache when valid

The validation table reports dataset status, file names, row counts, and missing columns.

Coordinate Reference System

Each project can store coordinate reference system metadata in project.json. The project CRS is the working CRS used by plan, section, and desurveyed interval geometry.

New projects default to Unknown CRS. During setup or in Project Settings, choose source and target presets such as:

- WGS84 longitude/latitude (EPSG:4326)
- NAD83 / BC Albers (EPSG:3005)
- NAD83(CSRs) / BC Albers (EPSG:3153)
- NAD83 / Statistics Canada Lambert (EPSG:3347)
- NAD83(CSRs) / Statistics Canada Lambert (EPSG:3348)
- NAD83 / Canada Atlas Lambert (EPSG:3978)
- NAD83(CSRs) / Canada Atlas Lambert (EPSG:3979)

- North America WGS84 / UTM zones 7N-22N (EPSG:32607 to EPSG:32622)
- North America NAD83 / UTM zones 7N-22N (EPSG:26907 to EPSG:26922)
- NAD83(CSRS) / UTM zones 7N-22N
- North America NAD27 / UTM zones 7N-22N (EPSG:26707 to EPSG:26722)
- NAD83 longitude/latitude (EPSG:4269)
- NAD27 longitude/latitude (EPSG:4267)
- NAD83 / CONUS Albers (EPSG:5070)
- NAD83 / Alaska Albers (EPSG:3338)
- US National Atlas Equal Area (EPSG:2163)
- North America WGS84, NAD83, and NAD27 UTM zones 1N-23N for USA/Alaska/Pacific/Caribbean coverage
- PNG94 longitude/latitude (EPSG:5546)
- PNG94 / PNGMG94 zones 54-58 (EPSG:5550, EPSG:5551, EPSG:5552, EPSG:9874, EPSG:9875)
- WGS84 / UTM zones 54S-58S (EPSG:32754 to EPSG:32758)
- Africa WGS84 / UTM zones 28N-39N (EPSG:32628 to EPSG:32639)
- Africa WGS84 / UTM zones 28S-39S (EPSG:32728 to EPSG:32739)
- Arc 1950 / UTM zones 34S-37S (EPSG:20934 to EPSG:20937)
- Hartebeesthoek94 / Lo15-Lo33 (EPSG:2046 to EPSG:2055)
- ETRS89 longitude/latitude (EPSG:4258)
- Europe ETRS89 / UTM zones 28N-38N (EPSG:25828 to EPSG:25838)
- ETRS89 / LAEA Europe (EPSG:3035)
- ETRS89 / LCC Europe (EPSG:3034)
- SWEREF99 TM - Sweden (EPSG:3006)
- EUREF-FIN / TM35FIN(E,N) - Finland (EPSG:3067)
- ETRS89 / DKTM1-DKTM4 - Denmark (EPSG:4093 to EPSG:4096)
- ISN93 / Lambert 1993 and ISN2004 / Lambert 2004 - Iceland (EPSG:3057, EPSG:5325)
- OSGB36 / British National Grid (EPSG:27700)
- IRENET95 / Irish Transverse Mercator (EPSG:2157)
- RGF93 v1 / Lambert-93 - France (EPSG:2154)
- Amersfoort / RD New - Netherlands (EPSG:28992)
- Asia WGS84 / UTM zones 40N-54N (EPSG:32640 to EPSG:32654)
- Asia WGS84 / UTM zones 46S-53S (EPSG:32746 to EPSG:32753)
- CGCS2000 longitude/latitude - China (EPSG:4490)
- JGD2011 longitude/latitude and UTM zones 51N-56N - Japan (EPSG:6668, EPSG:6688 to EPSG:6693)
- KGD2002 / Unified CS - South Korea (EPSG:5179)
- VN-2000 / UTM zones 48N-49N - Vietnam (EPSG:3405, EPSG:3406)
- DGN95 / UTM zones 46N-54N and 46S-54S - Indonesia (EPSG:23866 to EPSG:23874, EPSG:23876 to EPSG:23884)
- GDA2020 / MGA zones 49-56 (EPSG:7849 to EPSG:7856)
- GDA94 / MGA zones 49-56 (EPSG:28349 to EPSG:28356)
- AGD66 / AMG zones 49-56 (EPSG:20249 to EPSG:20256)
- AGD84 / AMG zones 49-56 (EPSG:20349 to EPSG:20356)
- Custom EPSG
- Custom/local grid

The source CRS describes the collar coordinates in the input files. The target/project CRS describes the coordinates used inside GeoChem Explorer after loading.

If Transform On Load is enabled and both source and target CRS are defined, the loader transforms collar X/Y coordinates into the target CRS before building plan, section, downhole, custom interval, and custom point geometry. Source CSV/XLSX files are not edited. Original source coordinate values are preserved in loaded rows as `source_<column>` fields.

Old Australian datum transforms such as AGD66 or AGD84 to GDA2020, Canadian and USA NAD27/NAD83/NAD83(CSRS) realization transforms, PNG94/WGS84 transforms, African Arc 1950 or Hartebeesthoek94 transforms, and European OSGB36/RD New/Iceland local datum transforms use approximate datum parameters in this version. GeoChem Explorer warns when no NTV2/grid-shift or time-dependent correction is applied. Treat these transforms as exploration-grade rather than survey-grade.

If CRS is unknown, the unit is not metres, collar coordinates look like degrees, transform is disabled, source and target are the same, or survey azimuths are marked magnetic, the loader warns but does not block the project.

Auto-Setup Project

Use Auto-Setup Project when creating a new project or when `project.json` does not exist yet.

The setup wizard:

1. Scans available CSV/XLSX files.
2. Infers which file is collar, survey, assay, lithology, geology, structural, and magnetic susceptibility.
3. Detects surface sample files such as soils or rock-chip point geochemistry.
4. Infers column mappings.
5. Lets you review and edit the mapping.
6. Detects sample registry assay joins where lab files contain only sample IDs.
7. Lets you add secondary assay sources.
8. Lets you add custom interval or point datasets.
9. Lets you choose an input source folder for future sync.
10. Lets you choose core photo library roots.
11. Writes `project.json`.
12. Loads the project.

Sample registry assay joins support projects where lab certificate files contain Sample or SAMPLE IDs, while hole ID and from/to intervals live in a separate sample register sheet. The loader joins those rows during project load, keeps duplicate or re-assay rows, and records source file and batch metadata in the working assay table.

If no downhole survey file exists but collar rows contain azimuth and dip, setup can enable a planned survey fallback. This creates one depth-0 survey row per collar and warns that the values are collar/planned orientations, not measured downhole survey stations.

Surface Samples

Surface sample support is for point-style geochemistry such as soils, rock chips, grab samples, auger points, or other sample grids that do not have true drillhole traces.

The app auto-detects these common file names in the project input folder:

- `surface_samples.csv`
- `SurfaceSamples.csv`
- `surface.csv`
- `soils.csv`

Auto-setup also scores files whose names include terms such as `surface`, `soil`, `soils`, `rockchip`, `rockchips`, `grab`, `sample`, or `samples`.

Required surface sample columns:

- Sample ID
- X
- Y

Common accepted Sample ID headings include `sample_id`, `sampleid`, `sample`, `Sample_ID`, `SampleID`, `Sample No`, and `SampleNo`. X/Y headings can be simple coordinate headings such as `x`, `y`, `easting`, `northing`, `east`, or `north`.

Optional surface sample columns:

- `rl`, `z`, or `elevation`
- `from`
- `to`
- `prospect`
- `dataset`, `sample_type`, or `Sample_Type`

All other numeric columns are treated as analytes and become available in the same analyte selectors and colour-scale tools as drillhole assays.

If `from` and `to` are present, the app uses them. If they are missing, the app defaults the surface sample interval to 0 to 0.1 m. This lets surface samples participate in the existing interval table without requiring users to invent downhole depths. Internally, each surface sample is translated into a synthetic collar, survey, and assay interval with `hole_type = SURFACE_SAMPLE`; in the UI it is shown separately as a surface sample, not as a drillhole.

Surface samples can be loaded in projects that also contain drillhole collar/survey/assay files, or in surface-only projects where the surface file provides the sample locations and assay values.

To add surface samples to an existing project:

1. Place the surface sample CSV in the project input folder using one of the common names above, such as `soils.csv` or `surface_samples.csv`.
2. Open Project Inputs.
3. Reload or run setup so the surface sample file is detected and mapped.
4. Review the validation table. Missing Sample ID, X, or Y values cause those rows to be dropped.
5. Open Plan View and use Layers to choose Both, Surface, or Drill Holes.

To colour surface samples in Plan View:

1. Set Layers to Surface or Both.

2. Leave Follow Drill Analyte on to colour surface samples by the active drill colour analyte, or turn it off and choose a separate Surface Analyte.
3. Use Surface Scale > Auto Jenks 6 for automatic natural-break colouring from the filtered surface values.
4. Use Surface Scale > Saved Preset to use a scale created in Analyte Colouring.
5. Use Surface Scale > Shared with Drill when drillholes are visible and both layers use the same analyte.

The sidebar keeps drillhole and surface sample ID filters separate. The Hole ID picker lists real drillholes only. The Surface Sample ID picker lists surface samples only.

Input Sync

Input Sync is designed for projects whose local input folder is copied from a shared source folder.

Controls:

- Choose Source Folder
- Preview Updates
- Update Inputs + Reload

When source file sizes change, the loader can ask whether to:

- Update From Source + Load
- Keep Local Edits + Load

Project Caches

The app writes project-local cache files under:

project-folder/.geochem-cache/

This can include loaded project cache, composites, saved subsets, saved date phases, lith colours, display scale presets, report snapshots, and structural stereonet cache.

Do not edit cache files unless you know exactly what you are changing.

14. project.json Essentials

project.json tells the app which files to load and how to map columns.

Main sections:

- files: input file names.
- coordinateReferenceSystem: project CRS metadata for display and validation warnings.
- columns: source column mappings.
- secondaryAssaySources: optional extra assay-like sources such as pXRF or extra lab sources.
- assaySampleJoins: optional sample registry joins for sample-only lab certificates.
- surveyFallback: optional collar planned-survey fallback when no survey file exists.
- customDatasets: optional interval or point data for display tracks.
- coreImages: core photo manifest paths or search roots.
- inputSync: source folder for input sync.
- reporting: report generator settings.
- calculatedVariables: configured ratios and atomic weights.

Default File Names

If not overridden, the app expects names such as:

- collar.csv
- survey.csv
- assays.csv
- surface_samples.csv
- lithology.csv
- geology.csv
- structural.csv
- magnetic_susceptibility.csv

Required Column Concepts

Collar:

- Hole ID
- X
- Y
- RL
- Optional prospect
- Optional completed date

Survey:

- Hole ID
- Depth
- Azimuth
- Dip

Assays:

- Hole ID
- From
- To
- Optional prospect
- Optional dataset
- Optional major split

Lithology:

- Hole ID
- From
- To
- Lith code

Geology:

- Hole ID
- From
- To
- Oxidation or weathering field

Structural:

- Hole ID
- Depth
- Structure type
- Alpha/beta/gamma or dip/dip-direction
- Orientation type
- Orientation quality

Magnetic susceptibility:

- Hole ID
- Depth
- Value
- Optional dataset

Surface samples:

- Sample ID
- X
- Y
- Optional RL/elevation
- Optional from/to interval
- Optional prospect
- Optional dataset or sample type
- Analyte columns

15. Common Workflows

Create a North/South Hole Subset from the Plan Map

1. Apply broad filters such as prospect and hole type.
2. In Plan mode, use Plotly rectangle or lasso selection to select the desired holes or intervals.
3. Click Use selected or visible holes.
4. Confirm the Hole ID filter now contains only the intended holes.
5. Open New Subset....
6. Check Current holes to save.
7. Enter a subset name and save.

If no intervals are selected when you click the button, it uses the current viewport instead.

Run PCA on a Filtered Dataset

1. Choose raw, 2m composite, or 5m composite interval mode.
2. Apply filters.
3. Open Tools > PCA.
4. Choose analytes with PCA Analytes....
5. Set components and optional row cap.
6. Click Run PCA.
7. Open Scatter panel PCA mode.
8. Inspect scores, colouring, and loadings.

Run Clustering and Inspect Results

1. Apply filters and choose a suitable interval mode.

2. Open Tools > Clustering.
3. Choose auto or manual K.
4. Select a cluster assay suite.
5. Click Run Clusters.
6. Use Scatter or Plan colour by cluster.
7. Use Downhole multi-track or cluster bars to inspect intervals.
8. Open View Profile for z-score/raw mean cluster summaries.
9. Use Assign Lith... to map cluster IDs to lith interpretations.

Build Downhole Intercepts

1. Open Tools > Downhole Intercepts.
2. Choose primary and optional secondary analytes.
3. Set cutoff, internal dilution, and minimum length.
4. Open the Downhole panel.
5. Choose or select a hole.
6. Click Generate intercepts/stats.

Export Data

Use the top-bar Export menu:

- Selected Intervals CSV
- Filtered Intervals CSV
- Filtered Hole List CSV

Filtered interval export requires the filtered interval set to be complete rather than a preview sample.

Display Soil or Surface Geochemistry Only

1. Add or load a surface sample file such as soils.csv.
2. Open Plan View.
3. Set Layers to Surface.
4. Turn off Follow Drill Analyte if you want a different surface analyte from the main drillhole colour analyte.
5. Choose Surface Analyte.
6. Choose Surface Scale.

Use Auto Jenks 6 for a quick statistically derived surface scale. Use Saved Preset for a company or project standard scale created in Analyte Colouring.

16. Notes and Limitations

- Large raw filters may enter preview mode. Some analysis tools such as PCA and clustering are disabled until the working set is exact.
- Optional panels and controls depend on project data. For example, stereonet structural mode needs structural rows, lith colouring needs lithology, and core photo tools need configured core photo manifests.
- The Report Generator currently targets the configured Reaper Executive EM reporting workflow.
- Project caches improve load speed but are derived artifacts; source CSV/XLSX files and project.json remain the authoritative project definition.